Nimitz (CVN 68) class (CVNM)

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Publication: Fighting Ships

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Contents

2021-11-19

Fleetlist

Ship totals	Nimitz
IN SERVICE	10

Name	*No*	*Builders*	*Laid down*	*Launched*	*Commissioned*	*F/S*
Nimitz	CVN 68	Newport News Shipbuilding	22 Jun 1968	13 May 1972	3 May 1975	PA
Dwight D Eisenhower	CVN 69	Newport News Shipbuilding	15 Aug 1970	11 Oct 1975	18 Oct 1977	AA
Carl Vinson	CVN 70	Newport News Shipbuilding	11 Oct 1975	15 Mar 1980	13 Mar 1982	PA
Theodore Roosevelt	CVN 71	Newport News Shipbuilding	13 Oct 1981	27 Oct 1984	25 Oct 1986	AA
Abraham Lincoln	CVN 72	Newport News Shipbuilding	3 Nov 1984	13 Feb 1988	11 Nov 1989	AA

George Washington	CVN 73	Newport News Shipbuilding	25 Aug 1986	21 Jul 1990	4 Jul 1992	PA
John C Stennis	CVN 74	Newport News Shipbuilding	13 Mar 1991	13 Nov 1993	9 Dec 1995	PA
Harry S Truman	CVN 75	Newport News Shipbuilding	29 Nov 1993	7 Sep 1996	25 Jul 1998	AA
Ronald Reagan	CVN 76	Newport News Shipbuilding	12 Feb 1998	4 Mar 2001	12 Jul 2003	PA
George H W Bush	CVN 77	Newport News Shipbuilding	6 Sep 2003	8 Oct 2006	10 Jan 2009	AA

2023-08-16

Specifications

	Nimitz
Dimensions and weights	
Displacement	
standard:	74,086 tonnes (72,915.9 (uk) t) (81,665.8 t (short)) (74,086,000 kg) (CVN 68–70) 75,160 tonnes (73,973.0 (uk) t) (82,849.7 t (short)) (75,160,000 kg) (CVN 71)
full load:	92,955 tonnes (91,486.9 (uk) t) (102,465.3 t (short)) (92,955,000 kg) (CVN 68–70) 97,933 tonnes (96,386.3 (uk) t) (107,952.7 t (short)) (97,933,000 kg) (CVN 71) 103,637 tonnes (102,000.2 (uk) t) (114,240.2 t (short)) (103,637,000 kg) (CVN 72–77)
Length	
overall:	332.9 m (1,092.2 ft) (CVN 68–77)

between perpendiculars:	317 m (1,040.0 ft) (CVN 68–77)
Beam	
overall:	76.8 m (252.0 ft) (CVN 68–77)
waterline:	40.8 m (133.9 ft) (CVN 68–77)
Draught	
hull:	11.3 m (37.1 ft) (CVN 68–70) 11.8 m (38.7 ft) (CVN 71) 11.9 m (39.0 ft) (CVN 72–76) 12.1 m (39.7 ft) (CVN 77)
Flight deck	
length overall:	332.9 m (1,092.2 ft) (CVN 68–77)
width overall:	76.8 m (252.0 ft) (CVN 68–77)
angled deck length:	237.7 m (779.9 ft) (CVN 68–77)
launch mechanism:	catapult
Aircraft lift	
number:	4
Performance	
Speed	
top speed:	30 kt (est) (55.6 km/h est) (34.5 mph est)
Capacity	
Complement	
military	

crew:	5,750
officers:	505
flag staff:	70
air group:	2,480
Machinery:	Nuclear; 2 Westinghouse/GE PWR A4W/A1G reactors; 4 turbines; 280,000 hp (209 MW); 4 emergency diesels; 10,720 hp (8 MW); 4 shafts; fp props
Firepower	
Missiles:	**SAM:** 2 Raytheon GMLS Mk 29 launchers [Ref 1]; Raytheon RIM-162D ESSM Block I; inertial guidance with mid-course update and semi-active radar terminal homing to 55 km (29.7 n miles) at Mach 3.6; warhead 39 kg. 2 Raytheon Mk 49 21-cell launchers [Ref 2]; 21 Raytheon RIM-116A/B RAM Block 0/I; passive IR/anti-radiation homing to 9.6 km (5.2 n miles) at Mach 2.5; warhead 9.3 kg.
Guns:	2 Raytheon Mk 15 Phalanx Block 1B with M61A1 20 mm/99; 6 barrels per mount; 4,500 rds/min combined to 1.5 km (0.8 n miles); on-mount search radar and EO/IR director. 4 (CVN 68, 70, 72, 76) Mk 38 Mod 2/3 with M242 25 mm/87 Bushmaster.
Physical countermeasures:	**Decoys:** SLQ-25 Torpedo Countermeasures Transmitting Set (Nixie). Mk 53 Nulka Mod 7 DLS; 6 Mk 137 Mod 10 2-barrelled fixed launchers; active RF decoys (CVN 69, 72, 73).
Electronic countermeasures:	**ESM/ECM:** SLQ-32(V)4 intercept and jammers.
Radars:	**Air search:** SPS-48E/G [Ref 3]; 3D; E/F-band. SPS-49A(V)1 [Ref 4]; C/D-band. SPQ-9B [Ref 5]; I/J-band. **Surface search:** SPS-67(V)1; G-band. **CCA:** SPN-41, SPN-43C, 2 SPN-46; E/F/J/K-band. TPX-42A Direct Altitude and Identity Readout (DAIR). **Navigation:** SPS-73(V)12; Furuno 900; I/J-band. **Fire control:** 4 Mk 9; I/J-band (2 per GMLS Mk 29 launcher). **Tacan:** URN-25.

Combat data systems:	Links 4A, 11, 16 and Satellite Tadil J. GCCS (M) satcomS; SSR-1 (CVN 68, 70, 71, 72, 74), WCS-3A (UHF DAMA) (CVN 69, 73, 76, 77); SRC-61 DMR (satcom) (CVN 68–72, 74–77); WSC-9 and CBSP (CVN 73), WSC-6 (SHF), WSC-8 (SHF), USC-38 (EHF), SSR-2A (GBS). SSDS Mk 2 USG-2 Cooperative Engagement System.
Weapon control systems:	4 Mk 9 Mod 1 TJS directors (part of the NSSMS Mk 57 SAM system).
Fixed-wing aircraft:	Composition of air-wing depends on mission and typically includes: 44 F/A-18E/F Hornet; 5 EA-18G Growler; 4 E-2C/2D Hawkeye, and CMV-22B Osprey.
Helicopters:	A mix of MH-60S, MH-60R, and HH-60H.

2023-08-16

Programmes

Programmes:	*Nimitz* was authorised in fiscal year 1967, *Dwight D Eisenhower* in FY 1970, *Carl Vinson* in FY 1974, *Theodore Roosevelt* in FY 1980, and *Abraham Lincoln* and *George Washington* in FY 1983. Construction contracts for *John C Stennis* and *Harry S Truman* were awarded in June 1988 and for *Ronald Reagan* in December 1994. Authorised in FY 1999, construction contract for *George H W Bush* awarded in January 2001.
Modernisation:	CVN 68 completed a three-year Refuelling and Complex Overhaul (RCOH) in 2001. RCOH of CVN 69 started in 2001 and completed in January 2005. RCOH of CVN 70 started in November 2005 and completed in July 2009. RCOH of CVN 71 started in 2009 and was completed in 2013. RCOH of CVN 72 started in March 2013 and completed in May 2017. This included modifications to receive the F-35C Lightning II. SSDS Mk 2 Mod 0 originally installed in CVN 68 (upgraded to Mk 2 Mod 1 in 2006). This includes fitting two RAM systems and SPQ-9B radar vice Mk 23 TAS. SSDS Mk 2 Mod 1 fitted to CVN 68, 69, 76, and 77. RAM systems replace one Mk 29 and all Phalanx launchers on CVN 68 and 69. CVN 74 similarly refitted during 2005 docking but retains upgraded CIWS (Phalanx) mounts as well. The SSDS upgrade package in CVN 73, 74, and 75 is known as the Capstone combat system upgrade. Capstone was installed in CVN 70, 71, and 72 during RCOH. Additional Capstone upgrades are planned for CVN 73 RCOH, which started in August 2017 and completes in 2023. CVN 74 began RCOH in May 2021, with completion due in 2025. Both CVN 69 and 77 are to be modified to operate the unmanned MQ-25A tanker aircraft, a development of the UCLASS programme, whose delivery into service is expected from 2026. CVN 70 received modifications to operate F-35C during incremental availability completed at Bremerton in August 2020. SPS-49 and SPS-48 air-search radars are to be replaced by Raytheon SPY-6(V)3 Enterprise Air Surveillance Radar

	(EASR), and SPN-43C with Saab SPN-50(V)1.
Structure:	Damage control measures include sides with approximately 2.5 in Kevlar plating over certain areas of side shell and box protection over magazine and machinery spaces. Aviation facilities include four lifts, two at the forward end of the flight deck, one to starboard abaft the island and one to port at the stern. There are four steam catapults (C13-1 [CVN 68–71], C13-2 [CVN 72–77]) and four (or three on CVN 76 and 77) Mk 7 Mod 3 arrester wires. Launch rate is one every 20 seconds. The hangar can hold less than half the full aircraft complement. Deckhead is 25.6 ft. Aviation fuel, 8,500 tonnes. Tactical Flag Command Centre for Flagship role. During RCOH, CVN 68 and 69 fitted with reshaped island (the mainmast has three yardarms to support more antennas). Major structural differences in CVN 76 and 77 include: a three-wire arresting system (to replace the four-wire system), an enlarged island structure which incorporates a bigger bridge, a three yardarm mainmast and the after mast (separate in previous ships), and an internal ordnance elevator. Other changes include a bulbous bow to reduce drag and a modified flight deck (angled deck increased by 0.1°) to allow the use of two catapults while aircraft land.
Operational:	Multimission role of ‘strike/ASW'. From CVN 70 onwards ships have an A/S control centre and A/S facilities; CVN 68 and 69 are backfitted. Endurance of 16 days for aviation fuel (steady flying) with greater than 1 million miles before nuclear reactor refuelling is required. Only one refuelling is required in the life of the ship. Ships' complements and air wings can be changed depending on the operational task. CVNs 69, 72, 75, and 77 based at Norfolk, VA. CVNs 70 and 71 based at San Diego, CA, and CVNs 68 and 74 at Bremerton, WA. CVN 76 arrived at its new homeport Yokosuka in November 2015. CVN 73 began RCOH at Newport News in August 2017, and CVN 74 in May 2021. First launch of X-47B UCAS-D was conducted by CVN 77 in May 2013. Plans to cancel the planned RCOH for CVN 75 and retire the ship early were announced in March 2019, however, these were halted soon after. CVN 68 has had life extended by 13 months to 2026. CVN 69 service life 2027 unless also life extended.

NIMITZ (US Navy)

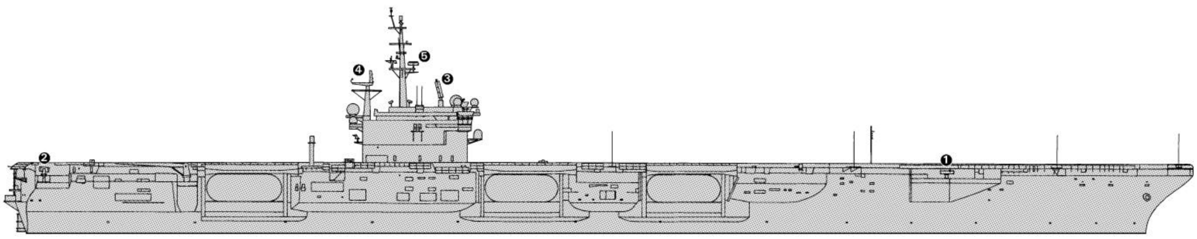
1787909

ABRAHAM LINCOLN (Tetsuya Kakitani)

2040764

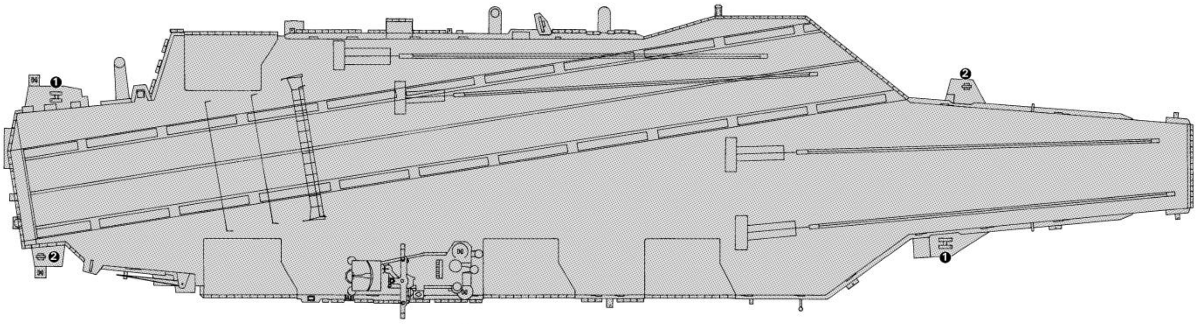
THEODORE ROOSEVELT (US Navy)

1764004



RONALD REAGAN (Ian Sturton)

1043489



RONALD REAGAN (Ian Sturton)

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2023-08-16

Additional Imagery

Online only images

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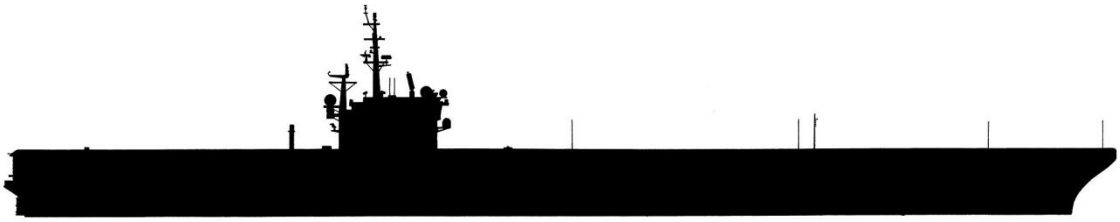

RONALD REAGAN SILHOUETTE (Ian Sturton)

1167586

GEORGE H W BUSH SILHOUETTE (Ian Sturton)

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